DOCUMENT PLAGIARISM DETECTION USING LSH

CIS 430 Project

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THE RESEARCH PROBLEM

- How do we detect plagiarism in thousands of documents?
- Current challenge: Must compare every document to every other
- For 10,000 documents = 50 MILLION comparisons
- This takes 14+ hours to run
- We need something faster

REAL-WORLD APPLICATIONS

Academic:

- University submissions
- Research paper screening
- Thesis verification

Industry:

- Google duplicate pages
- News article clustering
- Copyright detection

Scale: Need to process millions of documents efficiently

THE SCALABILITY CHALLENGE

Documents	Comparisons	Time
100	5,000	5 seconds
1,000	500,000	8 minutes
10,000	50,000,000	14 hours
100,000	5,000,000,000	58 days

This is $O(N^2)$ - NOT SCALABLE!

Proposed Solution: 3-Stage LSH Pipeline

Documents → Shingling → MinHashing → LSH → Candidates → (Sets) → (Signatures) → (Buckets) → (Verify)

Expected: $O(N^2) \rightarrow O(N)$ [Linear time!]

STAGE 1: Shingling (Converting Text to Sets)

Example with k=5:

Original text:
"a rose is a rose"

Shingles (5-character windows):
"a ros" | " rose" | "rose " | "ose i" | "se is"

Result: Each document = Set of overlapping text chunks

We hash these to numbers to save memory

Measuring Similarity: Jaccard Index

Formula: Jaccard = (Intersection) / (Union)

Example:
Doc A: {1, 2, 3, 4, 5}
Doc B: {3, 4, 5, 6, 7}

Shared shingles: {3, 4, 5} → 3
All shingles: {1,2,3,4,5,6,7} → 7

Jaccard = 3/7 = 43% similar

> 80% = Plagiarism 🚨

Stage 2: MinHashing (Creating Fingerprints)

Problem: Shingle sets are HUGE (1000s of elements)
Comparing them is still slow

Solution: Create small "signature" for each document

The Magic Property:
If two docs are 70% similar,
their MinHash signatures match 70% of the time!

Our Implementation:
- 100 hash functions
- Each document → 100 numbers (signature)
- Compression: 10,000 shingles → 100 numbers

Stage 3: LSH Banding (The Key Innovation)

Still $O(N^2)$ to compare all signatures...

LSH Solution:
1. Split signature into 20 bands × 5 rows
2. Hash each band independently
3. If ANY band matches → Candidate pair!

Why this works:
- Similar docs: Likely to match in ≥1 band
- Different docs: Unlikely to match in any band

Only verify the small set of candidates!

LSH Detection Probabilities

Similarity	Detection Chance	Verdict
90%	99.97%	✓ Caught
80%	98.5%	✓ Caught
70%	82.3%	✓ Caught
50%	12.4%	Ignored
30%	0.3%	Ignored

Parameters (b=20, r=5) tune this threshold

COMPLETE WORKFLOW

Phase 1: Preprocessing (one-time)

1. Shingle all documents
2. Build MinHash signature matrix
3. Apply LSH banding

Phase 2: Detection (fast!)

4. For each candidate pair:
   - Calculate true Jaccard similarity
   - If ≥ 80% → Flag as plagiarism

Complexity: $O(N)$ instead of $O(N^2)$

Dataset: 20 Newsgroups

Source:

- scikit-learn built-in
- ~20,000 newsgroup posts
- Well-known benchmark

Categories we'll use:

- Computer Graphics
- Medical Science
- Politics

Why this dataset?

✓ Easy to access
✓ Real-world text
✓ Diverse topics
✓ Natural duplicates
✓ 100-1000 words each

Expected: ~3,000 docs

How We'll Measure Success

Efficiency Metrics:
- Speedup: Naive vs LSH comparison count
- Runtime: Total time for detection

Accuracy Metrics:
- True Positives: Correctly found plagiarism
- False Positives: Incorrect flags
- Precision & Recall

Parameters to Test:
- Shingle size: k = 5, 7, 9
- Signature length: 50, 100, 200
- LSH bands: (20×5), (25×4), (10×10)

Expected Outcomes

METRIC	EXPECTED RESULT
Speedup	500-2000x faster
Pairs eliminated	99%+
Detection accuracy	**>95%** for high similarity
Runtime	<10 minutes total

Goal: Prove LSH scales to real-world document collections

SUMMARY & IMPACT

What We're Building:
✓ Complete LSH plagiarism detection system
✓ 500-2000x speedup over naive approach
✓ Tested on 3,000 real documents

Real-World Applications:
→ Billions of web pages
→ Social media duplicate detection
→ Academic paper screening

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bhumikabattineni@Bhumikas-MacBook-Pro-966 Project % source /Users/bhumikabattineni/Desktop/CIS430/Project/venv/bin/activate
(venv) bhumikabattineni@Bhumikas-MacBook-Pro-966 Project %  /usr/bin/env /Users/bhumikabattineni/Desktop/CIS430/Project/venv/bin/python
 /Users/bhumikabattineni/.vscode/extensions/ms-python.debugpy-2025.16.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launche
r 62221 -- /Users/bhumikabattineni/Desktop/CIS430/Project/lsh_detector.py
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DOCUMENT PLAGIARISM DETECTOR USING LSH
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[STEP 1] Loading 20 Newsgroups dataset...
Loaded 2903 documents in 0.65s

[STEP 2] Converting documents to shingles...
Shingling: 100%|██████████████████████████████████████████████████| 2903/2903 [00:00<00:00, 5017.04it/s]
Created shingle sets in 0.59s
Example: Doc 0 has 491 unique shingles

[STEP 3] Creating MinHash signatures...
MinHashing: 100%|═══════════════════════════════════════════════════| 2903/2903 [00:03<00:00, 858.61it/s]
Created signature matrix (100, 2903) in 3.38s

[STEP 4] Applying LSH to find candidate pairs...
LSH Banding: 100%|═══════════════════════════════════════════════════| 2903/2903 [00:00<00:00, 30105.69it/s]
LSH completed in 0.10s
Total possible pairs: 4,212,253
Candidate pairs found: 2,807
Reduction: 1500.6x fewer comparisons!

[STEP 5] Verifying candidate pairs...
Verifying: 100%|═══════════════════════════════════════════════════| 1000/1000 [00:00<00:00, 57110.43it/s]
Verification completed in 0.02s

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FINAL RESULTS
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Dataset: 2903 documents from 20 Newsgroups
Total possible comparisons: 4,212,253
LSH candidate pairs: 2,807
Speedup: 1500.6x

Plagiarism Detection (Jaccard ≥ 0.8):
True Positives: 17
False Positives: 983

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EXAMPLE OF DETECTED PLAGIARISM
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True Positives: 17
False Positives: 983

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EXAMPLE OF DETECTED PLAGIARISM
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Documents 1284 and 1455 are 91.7% similar

Document 1284 (first 300 chars):
Dear news readers,

Is there anyone using sheep models for cardiac research, specifically
concerned with arrhythmias, pacing or defibrillation? I would like
to hear from you.

Many thanks,

Andrew Mears
************** PLEASE EMAIL ME ************...

Document 1455 (first 300 chars):
Dear news readers,

Is there anyone using sheep models for cardiac research, specifically
concerned with arrhythmias, pacing or defibrillation? I would like
to hear from you.

Many thanks,
Andrew Mears...

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PROJECT COMPLETE!
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